Infobits

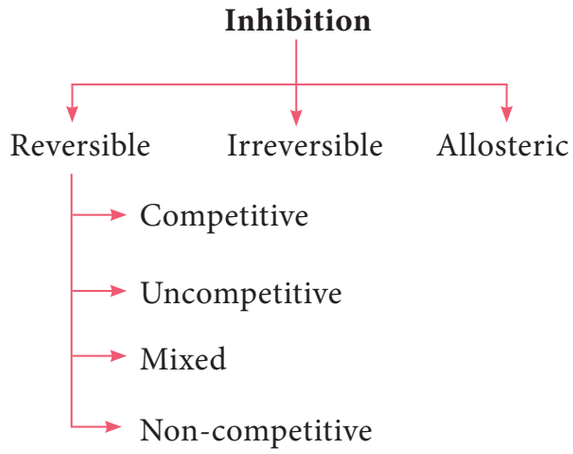
Idoenella sakaiensis is a bacterium capable of breaking down PET plastics. The bacterium first uses PETase to break down the PET plastic. This has potential importance in the recycling process of PET plastics.



Lipase is used in the determination of triglyceride and blood cholesterol level. Lipase producing microorganism have been found in industrial wastes, vegetable oil processing factories, diary plants and soil contaminated with oil.

4.10.4 Enzyme Regulation

Inhibitors: An enzyme inhibitor is a molecule that binds to an enzyme and decreases its enzyme catalyzing activity (Flowchart 4.1). This adverse affect of inhibitors on the rate of enzymatically catalyzed reactions are called inhibition.



Flowchart 4.1: Types of Inhibition

An irreversible inhibitor inactivates an enzyme by bonding covalently to a particular group at the active site. A reversible inhibitor inactivates an enzyme by non covalent, more easily reversible interactions. Competitive inhibitor is any compound that bears a structural resemblance to a particular substrate for binding at the active site of an enzyme. Non competitive inhibitors do not compete with the substrate for the enzyme's active site; instead, they interact with another part of the enzyme. Uncompetitive inhibitors bind only to the enzyme substrate complex without binding to the free enzyme (Figure 4.13).

a. Competitive Inhibition

b. Non-competitive Inhibition

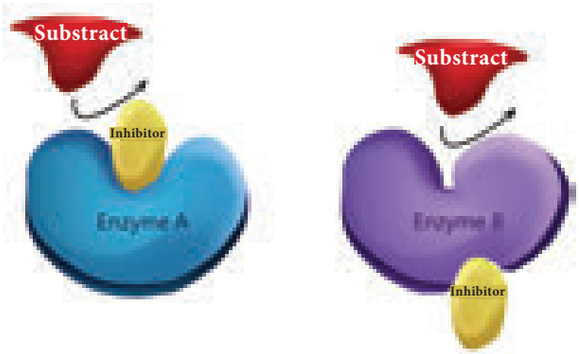


Figure 4.13: Competitive and non-competitive inhibition



Administration of the enzyme DNase I to the lungs of cystic fibrosis patients decrease the viscosity of the mucus and the breathing is made easier.

Feedback inhibition

In Feedback inhibition, the final product allosterically inhibits the enzyme that catalyses the first stage in the series

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